betaMC: Monte Carlo for Regression Effect Sizes

Ivan Jacob Agaloos Pesigan

Description

Generates Monte Carlo confidence intervals for standardized regression coefficients (beta) and other effect sizes, including multiple correlation, semipartial correlations, improvement in R-squared, squared partial correlations, and differences in standardized regression coefficients, for models fitted by `lm()`. `betaMC` combines ideas from Monte Carlo confidence intervals for the indirect effect (**Pesigan-Cheung-2023**) and the sampling covariance matrix of regression coefficients (Dudgeon, 2017) to generate confidence intervals effect sizes in regression.

Installation

You can install the CRAN release of `betaMC` with:

```
install.packages("betaMC")
```

You can install the development version of `betaMC` from GitHub with:

```
if (!require("pak")) install.packages("pak")
pak::pkg_install("jeksterslab/betaMC")
```

More Information

See GitHub Pages for package documentation.

References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. https://doi.org/10.1007/s11336-017-9563-z

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/